

# PHISHING URL SECURITY AUDIT REPORT

Executive Cybersecurity Risk Assessment & Explainable AI (XAI) Diagnostic Audit

SECURITY AUDIT  
REF: PG-AUDIT-00368

## 1. REPORT INFORMATION

|                |                                                                                                                                                                 |               |                                           |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------------|
| Report Title:  | Phishing URL Security Audit Report                                                                                                                              | Scan ID:      | #368 (PG-SCAN-00368)                      |
| Date and Time: | August 20, 2026 at 16:21:56 UTC                                                                                                                                 | Domain:       | paypal-security-update.account-verify.xyz |
| Analyzed URL:  | <a href="http://paypal-security-update.account-verify.xyz/signin.php?token=928103">http://paypal-security-update.account-verify.xyz/signin.php?token=928103</a> | Model Engine: | Random Forest (XAI Calibrated)            |

## 2. FINAL VERDICT

■ CRITICAL PHISHING THREAT DETECTED RISK LEVEL: CRITICAL

This URL exhibits verified malicious deception patterns characteristic of active credential harvesting or brand impersonation.

|                        |                        |                           |                         |                          |
|------------------------|------------------------|---------------------------|-------------------------|--------------------------|
| PREDICTION<br>PHISHING | RISK LEVEL<br>CRITICAL | CONFIDENCE SCORE<br>99.8% | PHISHING PROB.<br>99.8% | LEGITIMATE PROB.<br>0.2% |
|------------------------|------------------------|---------------------------|-------------------------|--------------------------|

## 3. URL SECURITY ANALYSIS

| Security Characteristic | Diagnostic Status              | Security Characteristic  | Diagnostic Status                                                 |
|-------------------------|--------------------------------|--------------------------|-------------------------------------------------------------------|
| HTTPS                   | No (Insecure HTTP)             | Domain                   | paypal-security-update.account-verify.xyz                         |
| URL Length              | 72 characters (Normal)         | Domain Length            | 41 characters                                                     |
| Number of Subdomains    | 1 level(s)                     | Suspicious Keywords      | Yes (3 keywords: <b>signin</b> , <b>verify</b> , <b>account</b> ) |
| IP Address Usage        | No (DNS Host)                  | Suspicious URL Structure | Yes (Hyphenated Brand Spoofing)                                   |
| TLD Abuse Rating        | 80% Risk (High-Risk Abuse TLD) | Shannon Entropy          | 4.74 bits (High Randomness)                                       |

## 4. WHY THIS RESULT WAS GIVEN

### Main Factors Influencing Decision (5 Detected):

- **Use of HTTP / Insecure Transport:** URL uses unencrypted plaintext HTTP protocol instead of encrypted HTTPS.
- **Suspicious Keywords Detected:** Found 3 sensitive credential/security keyword(s) ('signin', 'verify', 'account') in the URL.
- **Hyphenated Brand Spoofing:** Domain structure contains deceptive brand combinations (combosquatting).
- **High-Risk Top-Level Domain:** Registered under a TLD frequently associated with spam and automated phishing campaigns.
- **Unusual URL Characteristics (Entropy 4.74):** Elevated randomness indicates algorithmic domain generation.

5. EXPLAINABLE AI SUMMARY

Concise mathematical summary of the top 3–5 features from Explainable AI (SHAP & LIME) influencing the model's decision:

| Rank | Influential Feature       | Observed Value | Risk Impact Direction   | Short Impact Explanation                                                                          |
|------|---------------------------|----------------|-------------------------|---------------------------------------------------------------------------------------------------|
| #1   | TLD Risk Factor           | 0.8            | +Risk (Pushed Phishing) | Registered under a high-risk Top-Level Domain frequently associated with spam and phishing.       |
| #2   | Subdomain Count           | 1              | -Risk (Pushed Safe)     | Clean, standard top-level domain hierarchy without excessive nesting.                             |
| #3   | HTTPS Protocol            | 0              | +Risk (Pushed Phishing) | Insecure HTTP protocol strongly elevates the risk of spoofing.                                    |
| #4   | Suspicious Keywords Count | 3              | +Risk (Pushed Phishing) | Contains 3 deceptive security/credential harvest keywords.                                        |
| #5   | Shannon Entropy           | 4.74           | +Risk (Pushed Phishing) | High character entropy (4.74) suggests algorithmic domain generation or obfuscated payload paths. |

6. SECURITY RECOMMENDATION

■ CRITICAL SECURITY DIRECTIVES (HIGH RISK / PHISHING DETECTED)

- **Do not enter username or password:** This URL exhibits credential harvesting patterns.
- **Do not provide personal or financial information:** Never enter credit card, banking, or identity data.
- **Do not download files from the URL:** Files hosted on this domain may contain malware or spyware.
- **Verify the domain using a trusted source:** Navigate to official services directly via known bookmarks or official search.
- **Close or report the suspicious page if appropriate:** Terminate active browser tabs and report this link to your security team.

7. SHORT SECURITY SUMMARY

|              |                                                                                                                                                                 |                     |                                                                     |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------|
| URL:         | <a href="http://paypal-security-update.account-verify.xyz/signin.php?token=928103">http://paypal-security-update.account-verify.xyz/signin.php?token=928103</a> | Risk:               | CRITICAL                                                            |
| Verdict:     | PHISHING                                                                                                                                                        | Confidence:         | 99.8% (Phishing: 99.8%)                                             |
| Main Reason: | Detected 5 risk factors: Insecure HTTP, Credential keywords, Malicious pattern matching (Phishing Probability: 99.8%).                                          | Recommended Action: | DO NOT PROCEED. Do not enter passwords or submit sensitive details. |